

Previous attachment

TOSI x SUS x UAEM



Open Science Research Fair

The Bellini Atrium | Feb 20, 5-7pm

Explore the practice of open science in research, connect with PIs and graduate students, and enjoy refreshments!

Office hours

REVISED

- Fridays **12-1**pm (zoom or Stewart Biology, N5/8)

<https://mcgill.zoom.us/j/81407275167?pwd=8Cgm0RPnWQRLOdaCNFyXnS3M8amQUj.1>

- By appointment
- Before or after class, right here in Leacock 132

Large Scale Chromosomal Changes

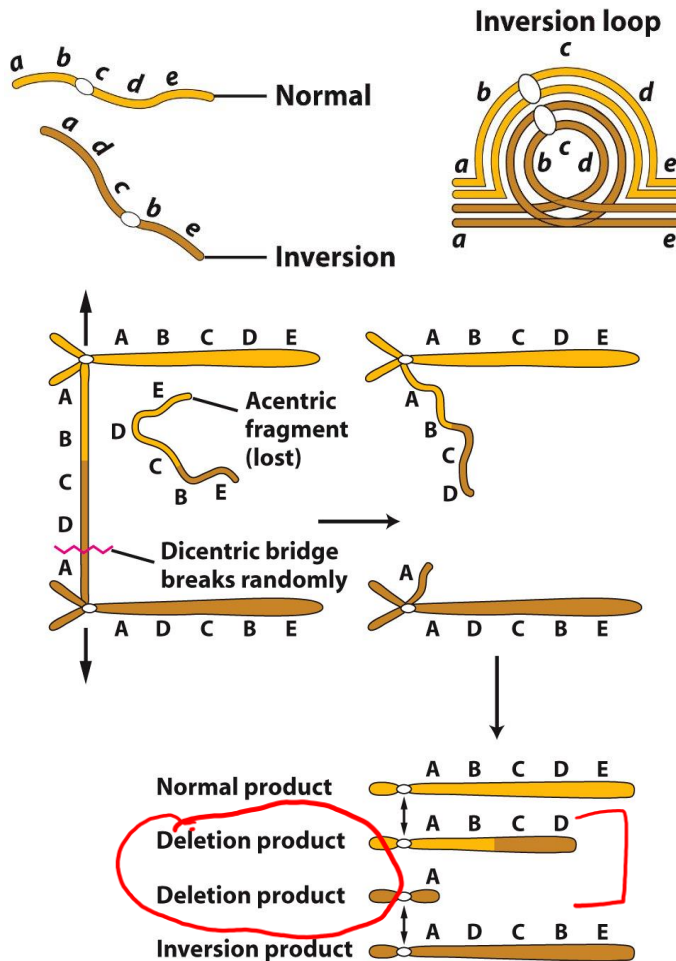
Changes in Chromosome Structure

Chromosomal inversions

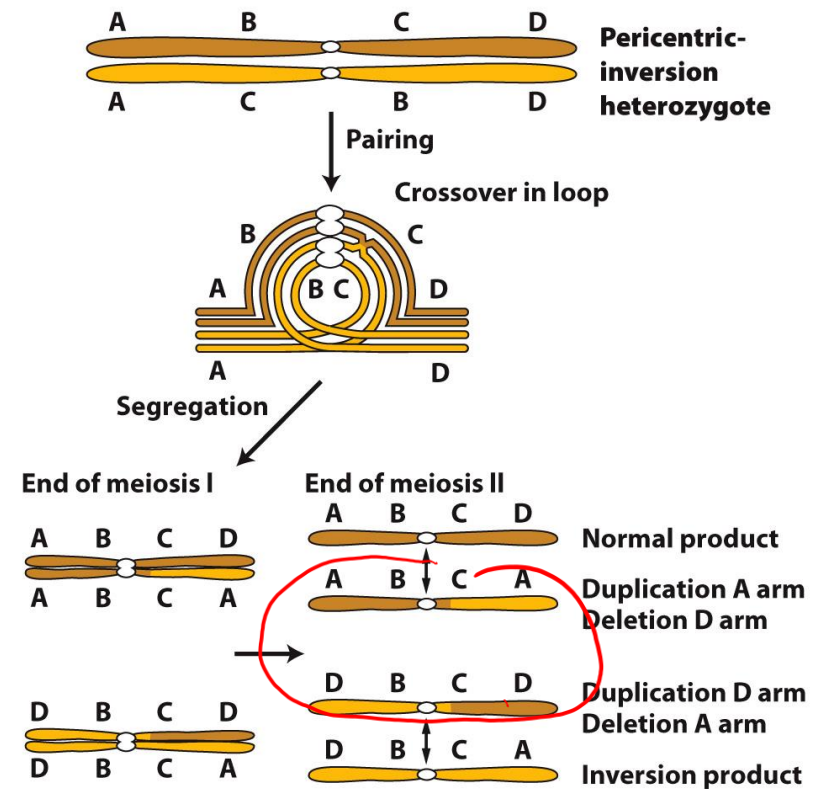
Effects on chromosome behavior during meiosis

Meiotic recombination in inversion heterozygotes leads to gametes with duplications and deletions

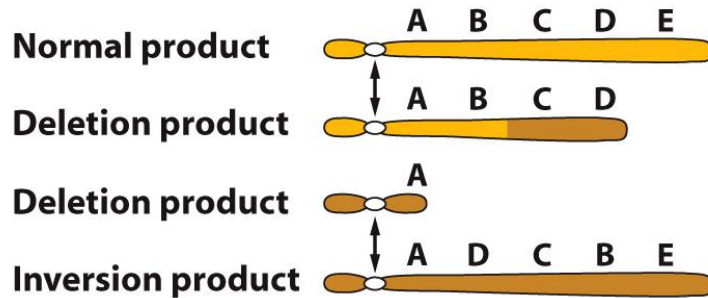
Paracentric inversion heterozygote



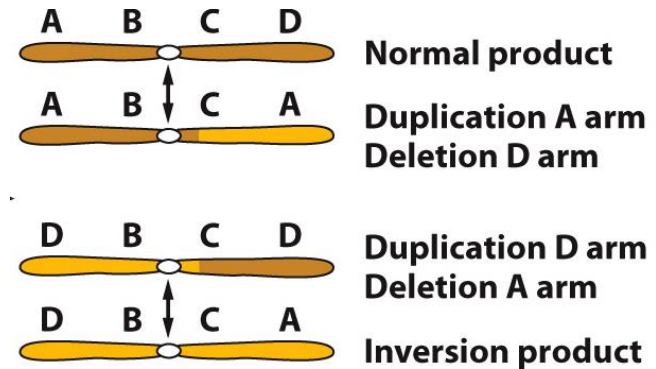
Pericentric inversion heterozygote



meiotic products: paracentric inversion



meiotic products: pericentric inversion



Think Break

What clinical consequences might you expect in an adult individual with a chromosomal inversion?

Effect on frequency of recombinant progeny

Example from *Drosophila*

dp: dumpy

cn: cinnabar

45 map units apart

$++/++$

x

$dp\ cn/dp\ cn$

$++/dp\ cn$

x

$dp\ cn/dp\ cn$

F1



F2

250

wild type

$++/dp\ cn$

246

dumpy
cinnabar

$dp\ cn/dp\ cn$

5

dumpy

$dp\ +/dp\ cn$

7

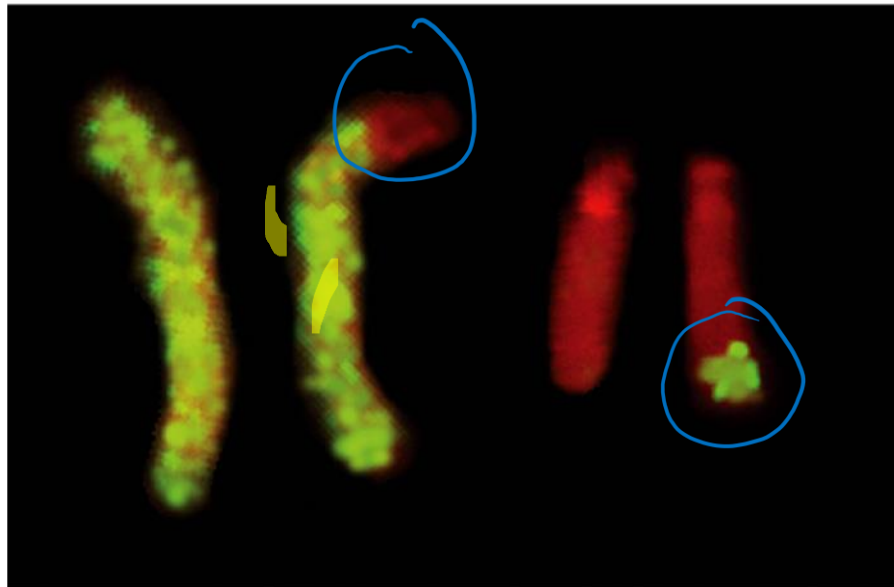
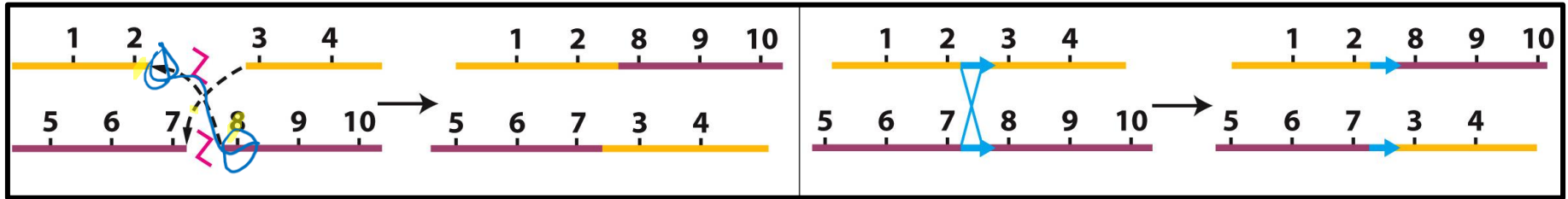
cinnabar

$+ cn/dp\ cn$

Expect 45%
recombinant
progeny

Germ cells with a
chromosome that
underwent crossing
over within the
inversion did not
produce viable
progeny.

Chromosomal translocations



Chapter 17 Opener

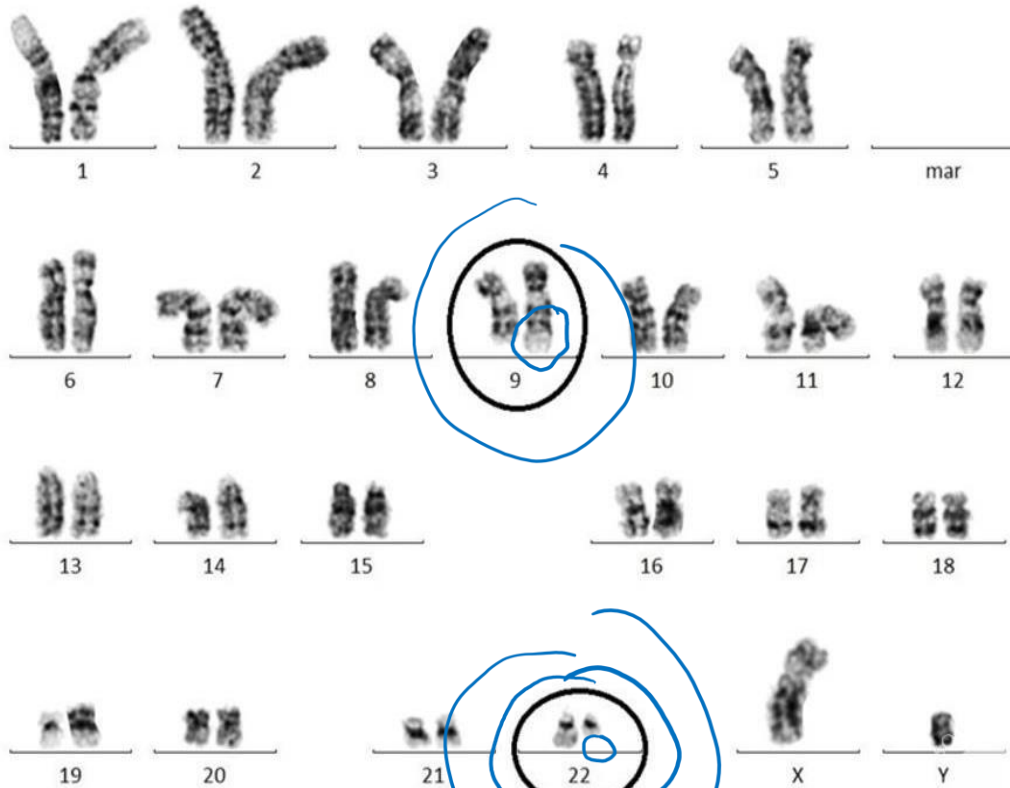
Introduction to Genetic Analysis, Eleventh Edition

Addenbrookes Hospital/Science Source

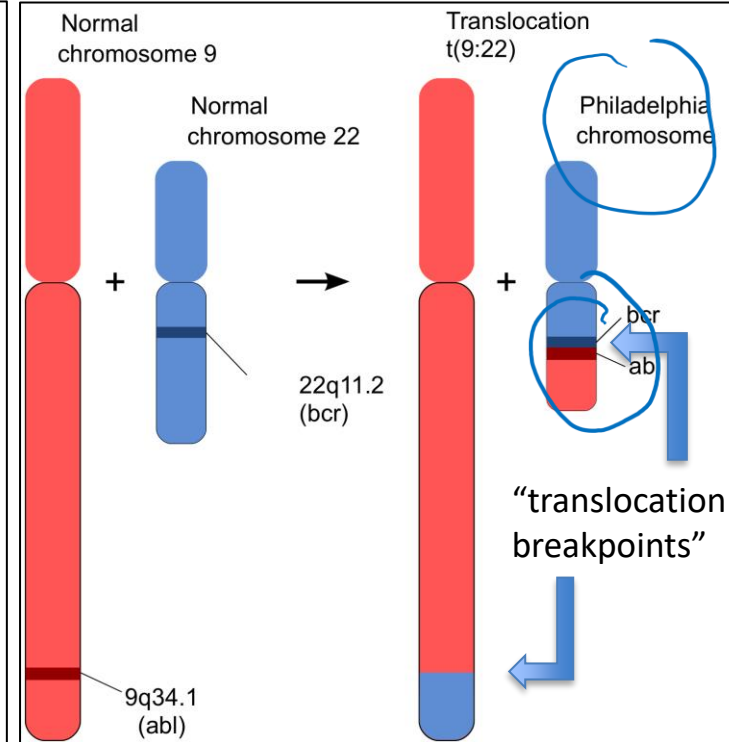
BIOL 202 W2025 | © LA Nilson

Chromosomal translocations

somatic consequences



First chromosome abnormality linked with a specific human cancer



Formation of a hybrid oncogene: chronic myelogenous leukemia

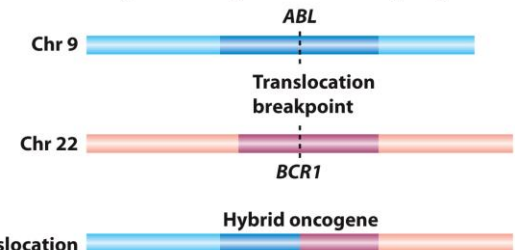
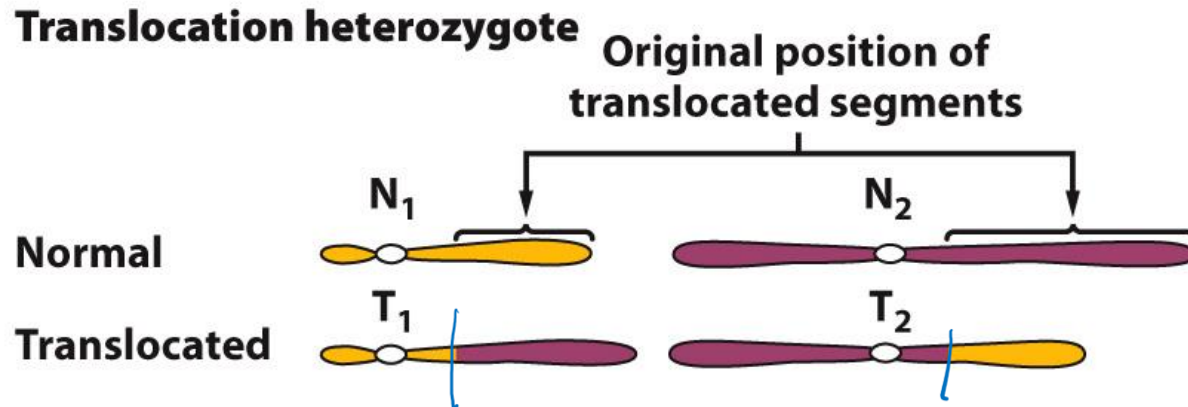


Figure 17-35b
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

Chromosomal translocations

somatic consequences

balanced translocations:

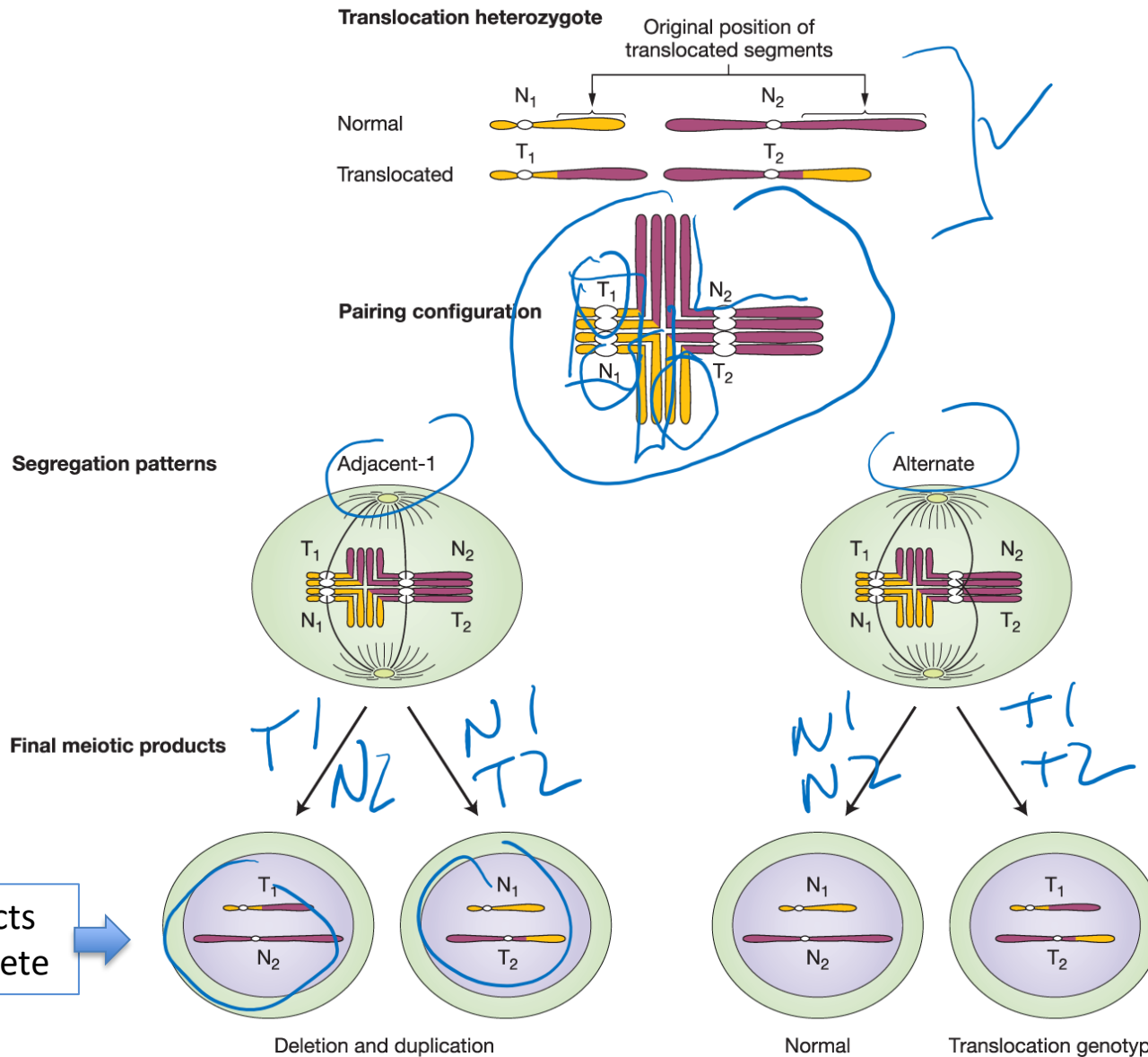


What phenotype would you predict for an individual with this translocation?

Somewhat "normal"

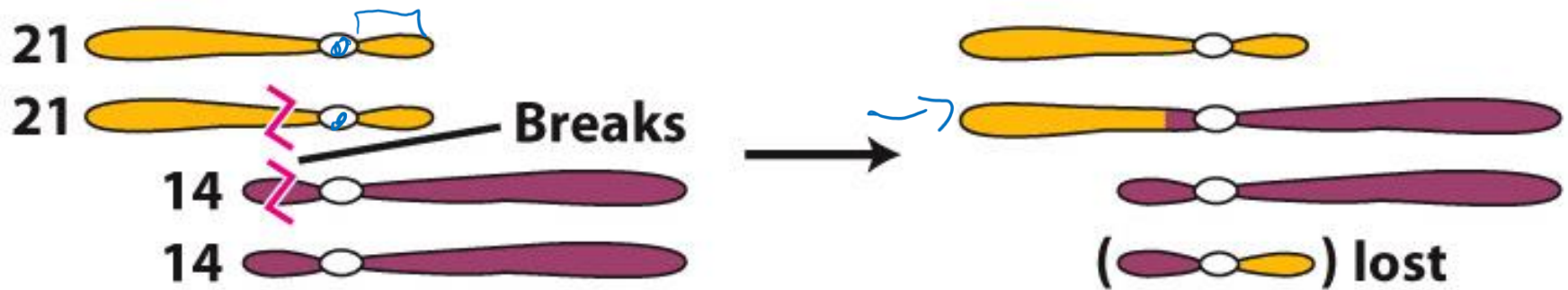
Balanced translocations and meiosis

two segregation patterns in a reciprocal-translocation heterozygote



Consequences of balanced translocations

Robertsonian translocations

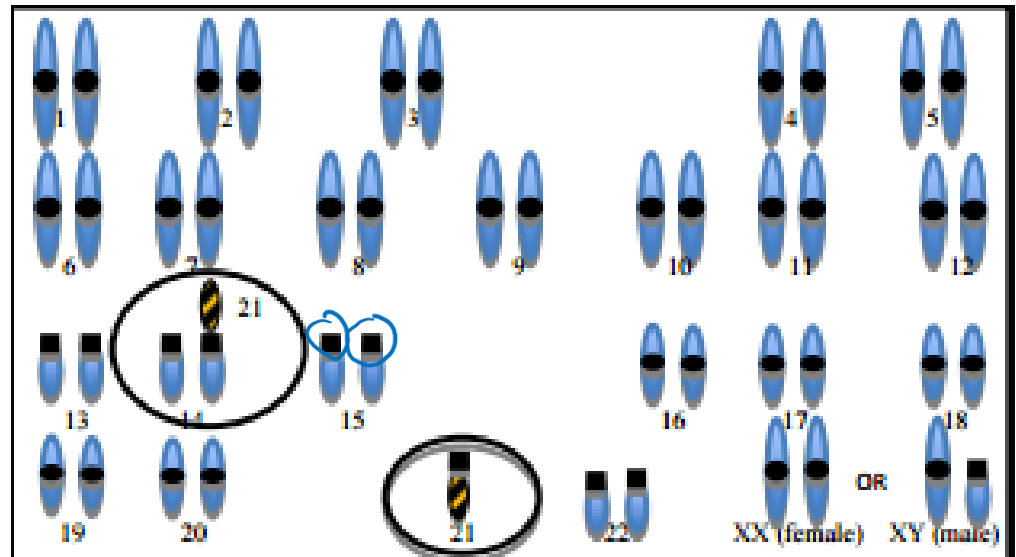


A type of balanced translocation

Can involve any of the acrocentric chromosomes: 13, 14, 15, 21, or 22

Frequency: 1/1000 babies

Phenotype usually normal

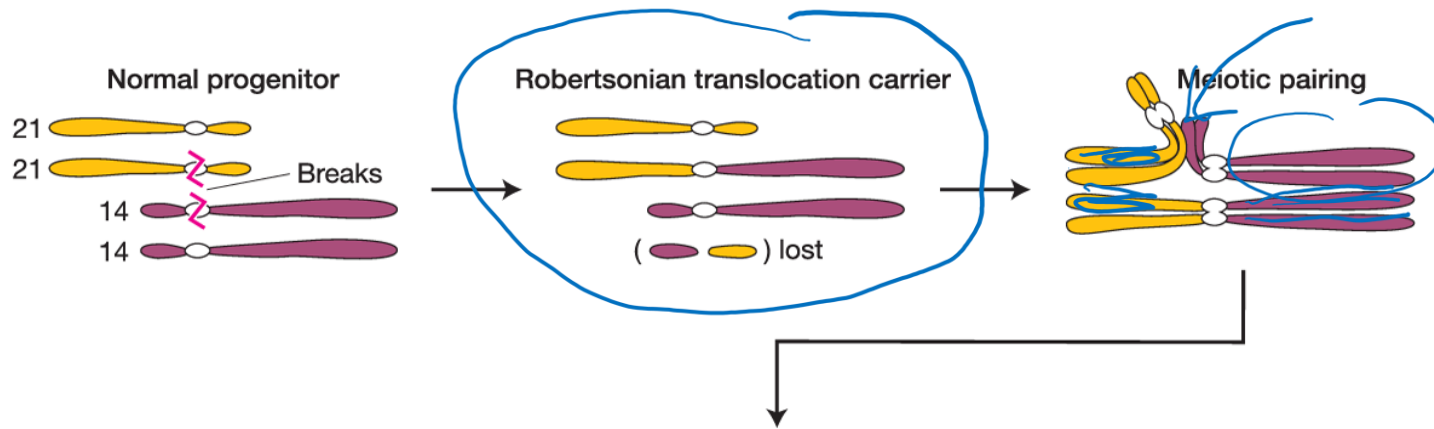




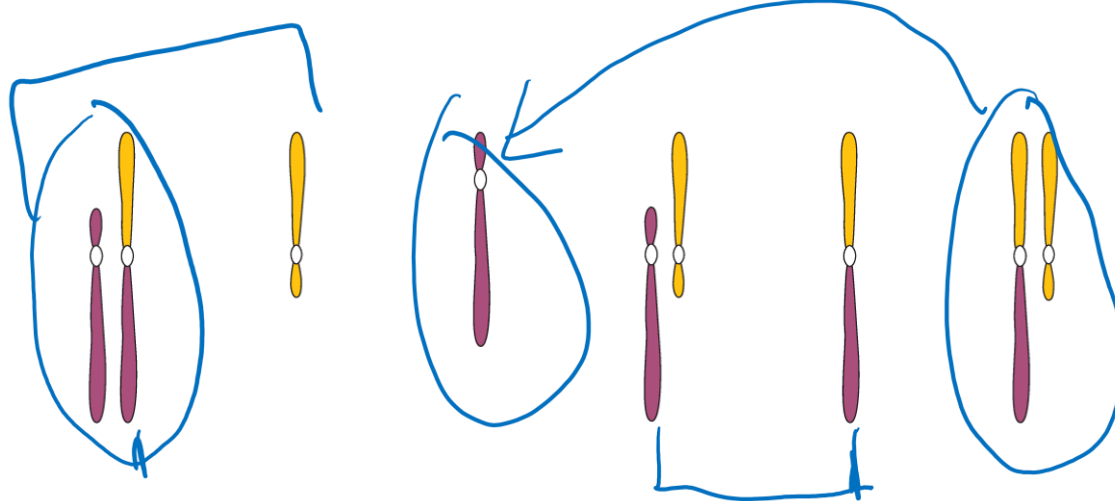
Think Break

- any genes on short arms are ^{help} sufficient
- break points not disrupting important regions

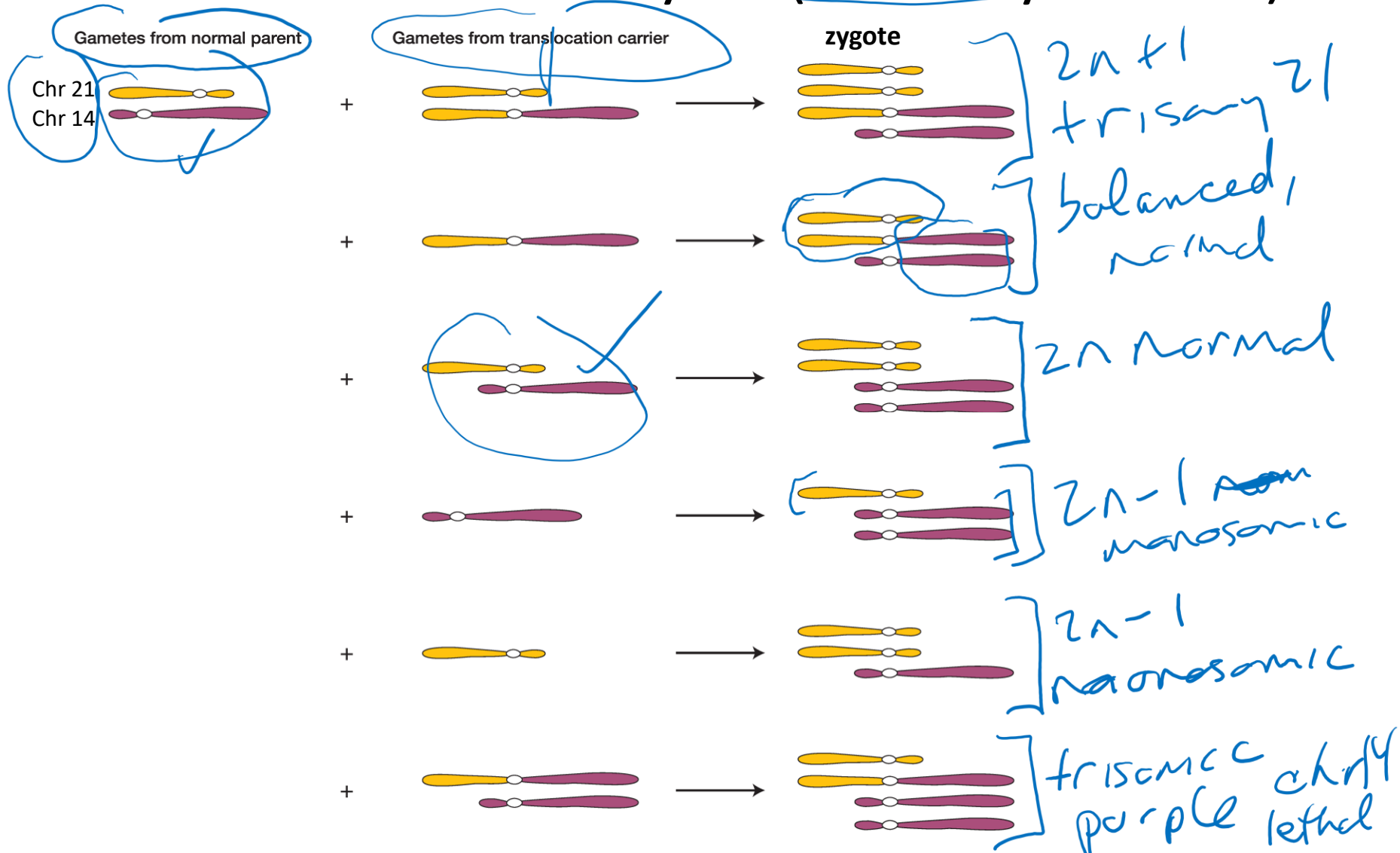
A balanced Robertsonian translocation resulting in inheritance of trisomy 21 (Down's Syndrome)



Segregation during meiosis → 6 possible gametes

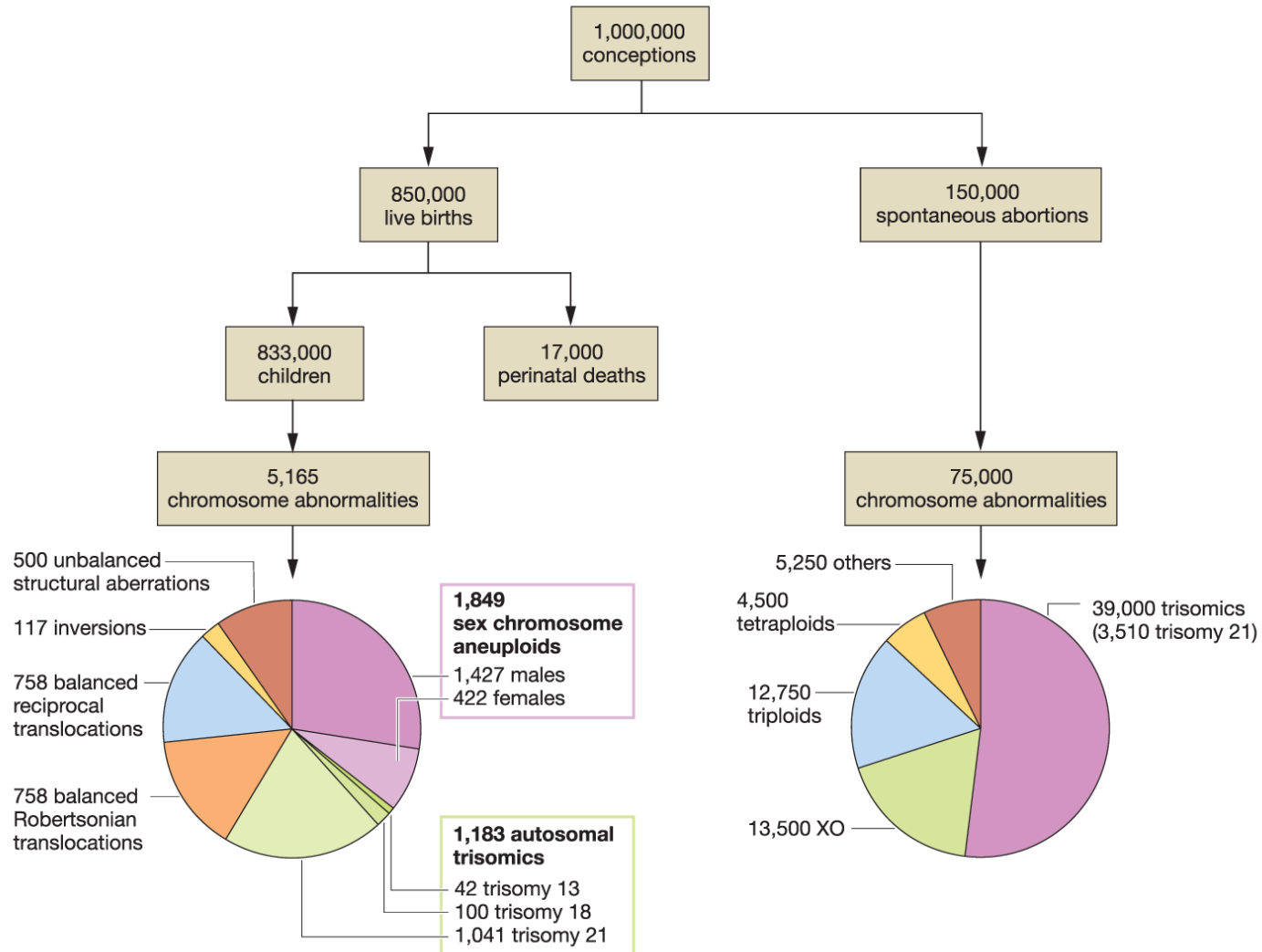


A balanced Robertsonian translocation resulting in inheritance of trisomy 21 (Down Syndrome)



The fates of a million implanted human zygotes

The fates of a million implanted human zygotes



Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

The Dynamic Genome

- Discovery of transposable elements through study of “unstable” alleles
- Impact of transposable elements on the genome
- Genomic surveillance mechanisms that limit transposon mobility

Discovery of transposable elements through study of “unstable” alleles

Maize as a system for studying inheritance of traits

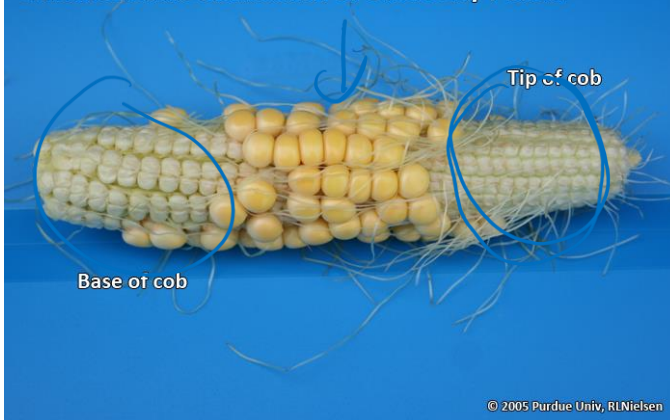
pigment genes



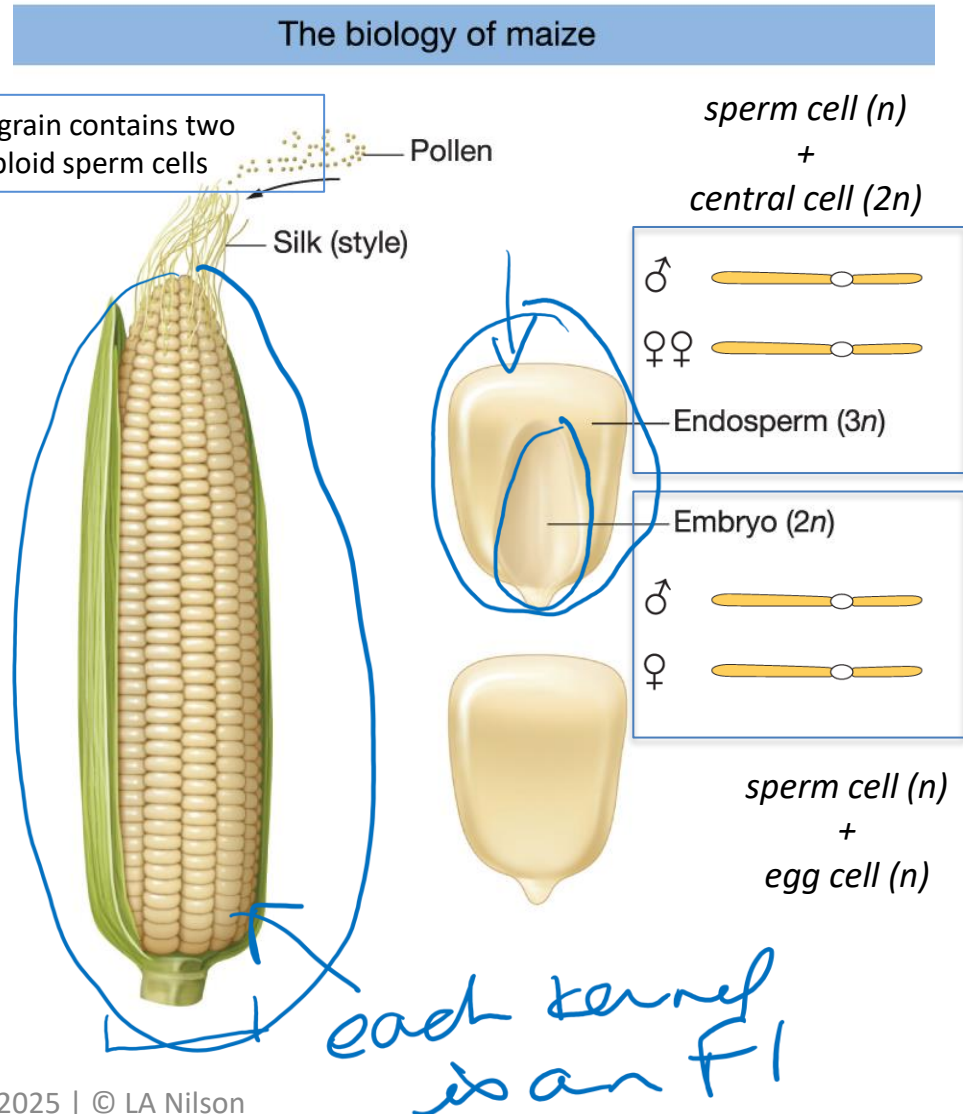
Department of Plant Breeding, Cornell University

Barbara McClintock and colleagues, 1929

Poor Kernel Set Due to
Unsuccessful Fertilization of Ovules by Pollen



© 2005 Purdue Univ, R.Nielsen

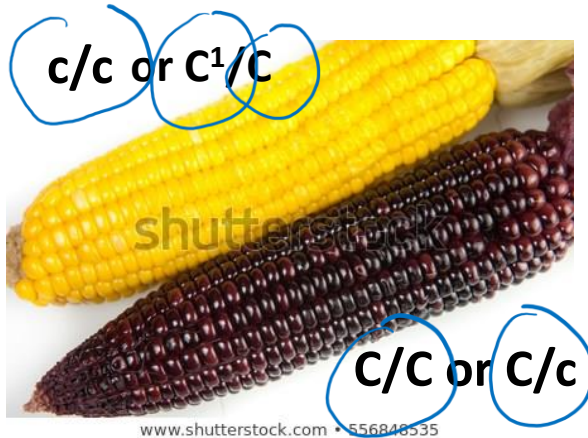


Maize as a system for studying inheritance of traits: pigment genes

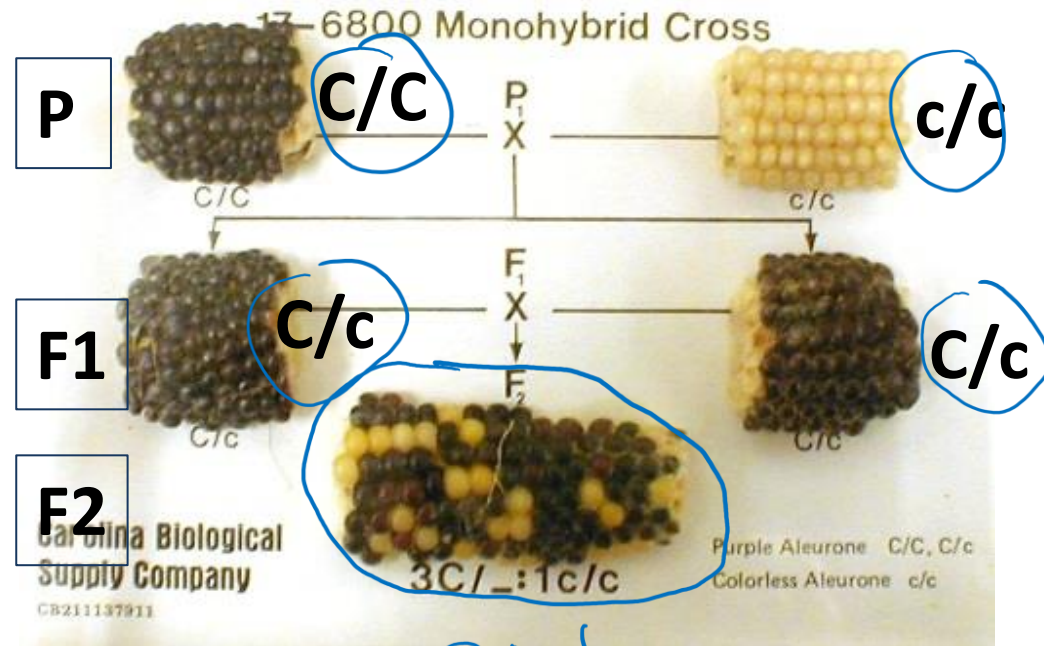
The *coloured aleurone* (C) gene is important for the production of purple pigment in maize.

c is a recessive allele that is unable to make pigment.

C^1 is a dominant inhibitor allele that represses pigment production



Inheritance of traits is easy to see



Normal 3:1 phenotypic ratio in the F₂

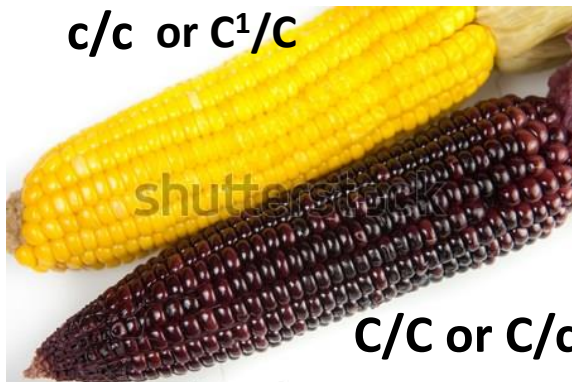
The order of dominance is $C^1 > C > c$.

Maize as a system for studying inheritance of traits pigment genes

The *coloured aleurone* (*C*) gene is important for the production of purple pigment in maize.

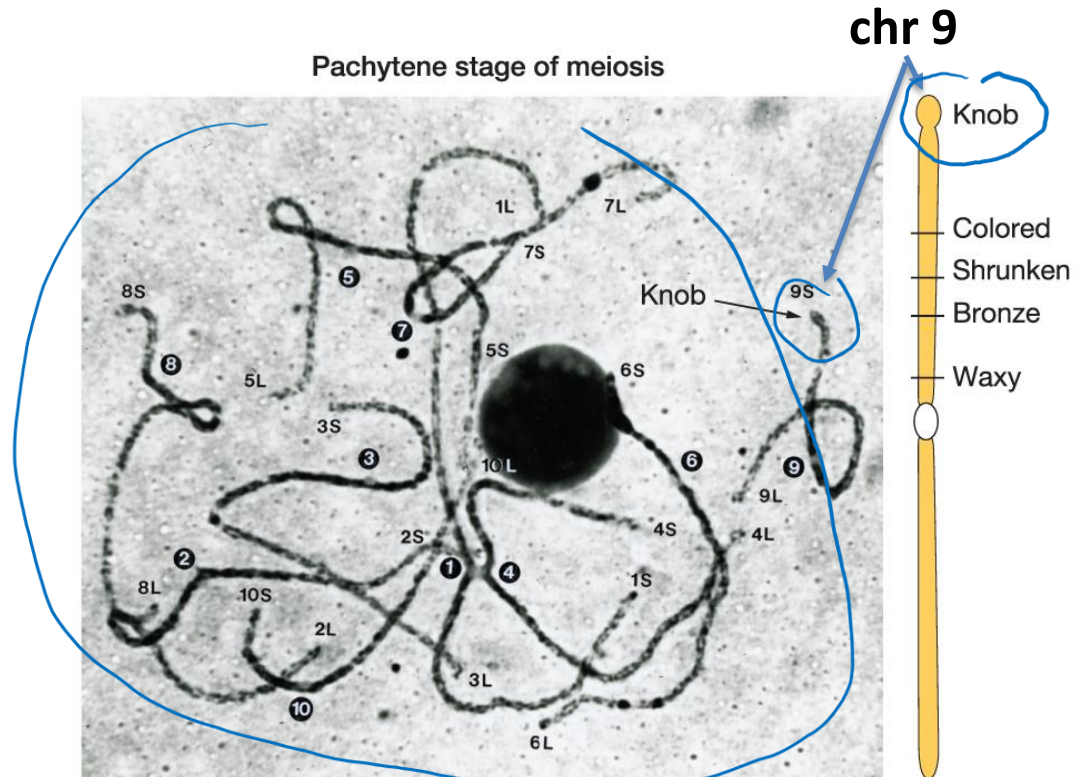
c is a recessive allele that is unable to make pigment.

*C*¹ is a dominant inhibitor allele that represses pigment production



The order of dominance is *C*¹ > *C* > *c*.

Established cytogenetic techniques



Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

Part (b) Photo kindly supplied by Ron Phillips, from a photo taken by John Stout. William Sheridan, University of North Dakota.